

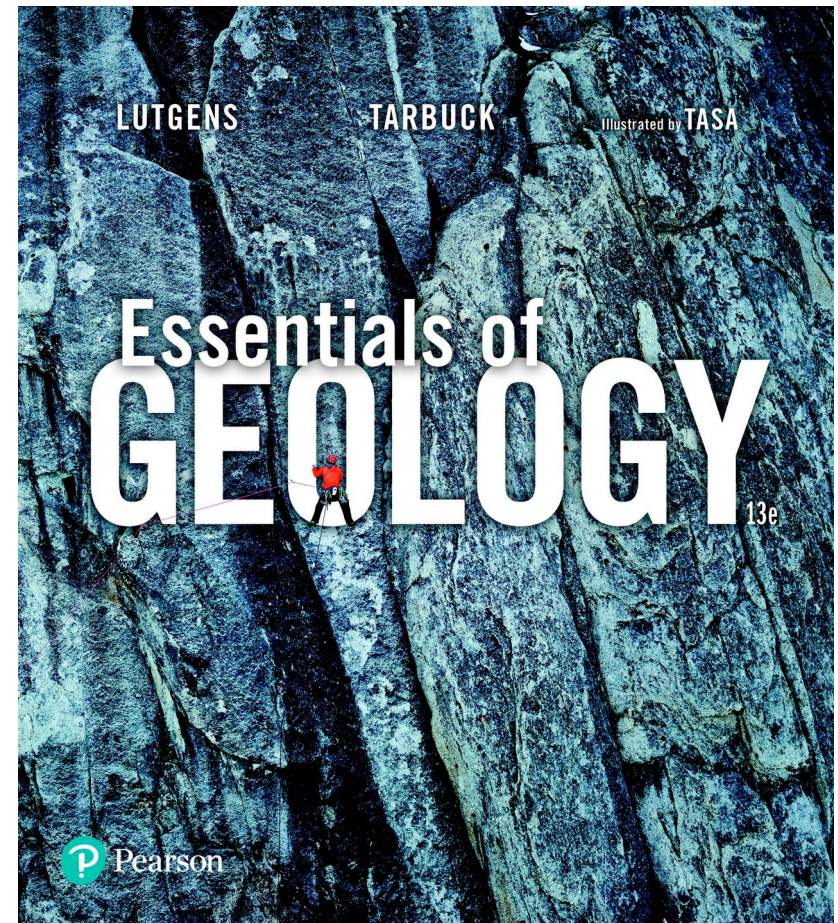
Lecture 10 Outline

Origin & Evolution of the Ocean Floor

Merry Wilson
Scottsdale Community College

Essentials of Geology

Thirteenth Edition



Learning Objectives

- **LO 10.1** Define *bathymetry* and describe the various bathymetric techniques used to map the ocean floor.
- **LO 10.2** Compare and contrast a passive continental margin with an active continental margin and list the major features of each.
- **LO 10.3** List and describe the major features of the deep-ocean basin.
- **LO 10.4** Sketch and label a cross-sectional view of the Mid-Atlantic Ridge. Explain how a cross section of the East Pacific Rise would look different.

Learning Objectives (cont.)

- **LO 10.5** Write a statement describing how spreading rates affect ridge topography.
- **LO 10.6** List the four layers of oceanic crust and explain how oceanic crust forms and how it differs from continental crust.
- **LO 10.7** Outline the steps by which continental rifting results in the formation of new ocean basins.
- **LO 10.8** Compare and contrast spontaneous subduction and forced subduction.

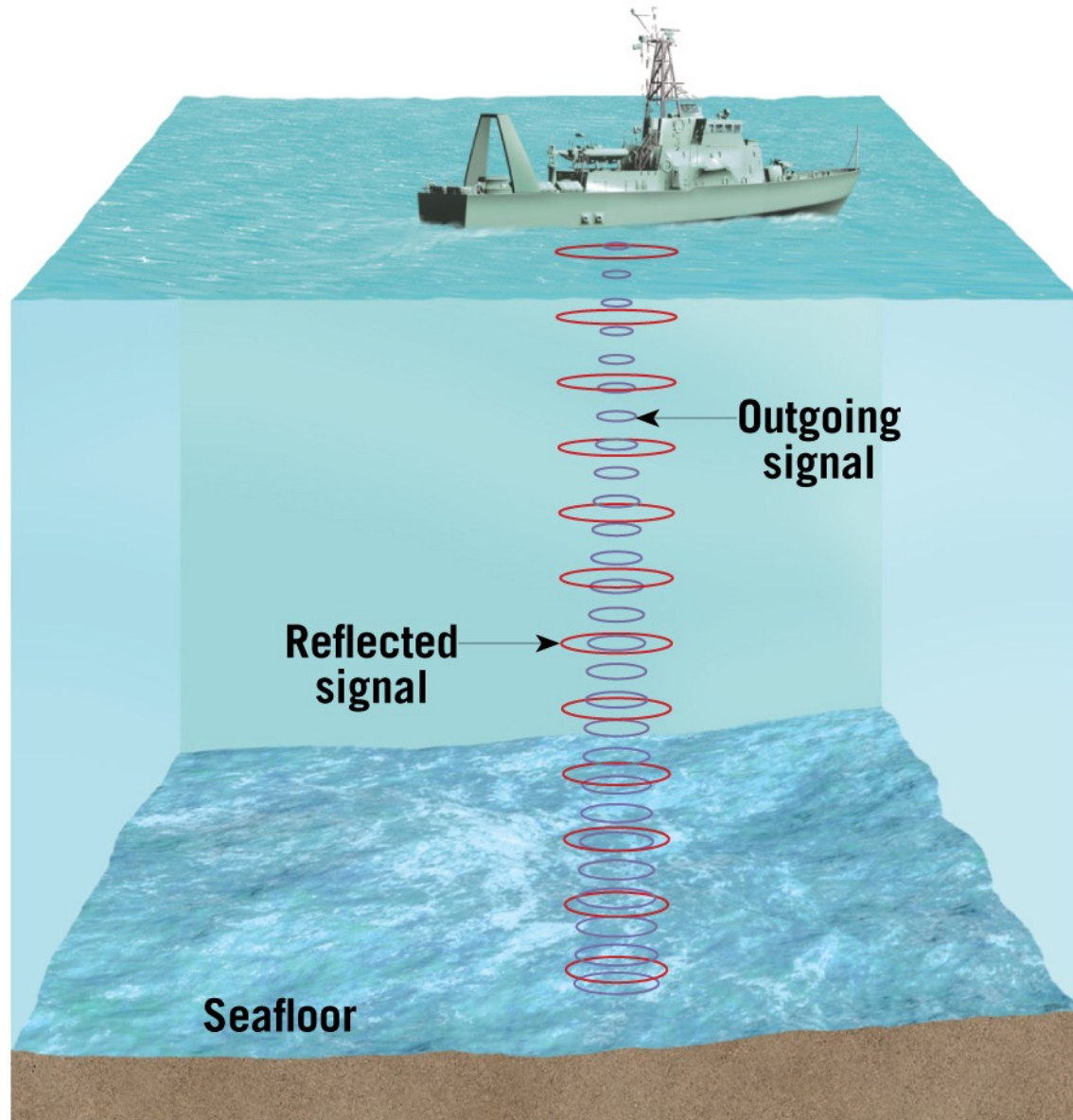
10.1 An Emerging Picture of the Ocean Floor

- Mapping the seafloor
 - From 1872 to 1876, the HMS *Challenger* collected oceanographic data.
 - Measured the depth to the seafloor by lowering weighted lines overboard.
 - Deepest spot measured is now called the *Challenger Deep*.

10.1 An Emerging Picture of the Ocean Floor (cont.)

- Mapping the Seafloor
 - Modern bathymetric techniques
 - The topography (shape) of the ocean floor is called bathymetry
 - Sonar, using sound energy, is now used to measure the depth to the ocean floor.
 - Early bathymetric profiles were created using echo sounders, which bounce a sound off an object to determine the distance.

Echo Sounder

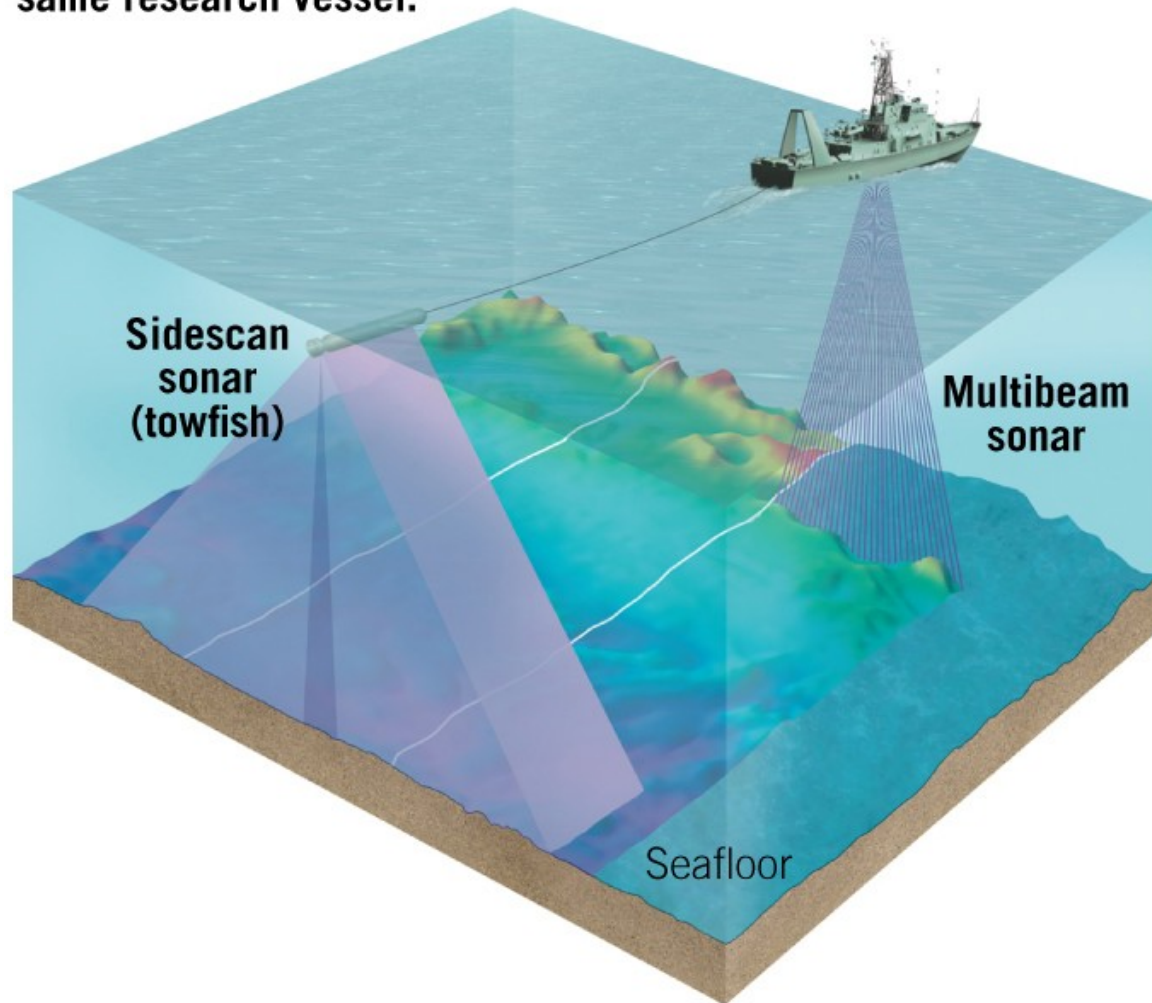


10.1 An Emerging Picture of the Ocean Floor

- Mapping the seafloor
 - Modern bathymetric techniques
 - Sidescan sonar images a horizontal region above the seafloor.
 - High-resolution multibeam instruments send out a fan of sound and record reflections from various receivers to provide a more detailed view of the ocean floor.
 - Only about 5 percent of the seafloor has been mapped in detail.

Sidescan and Multibeam Sonar

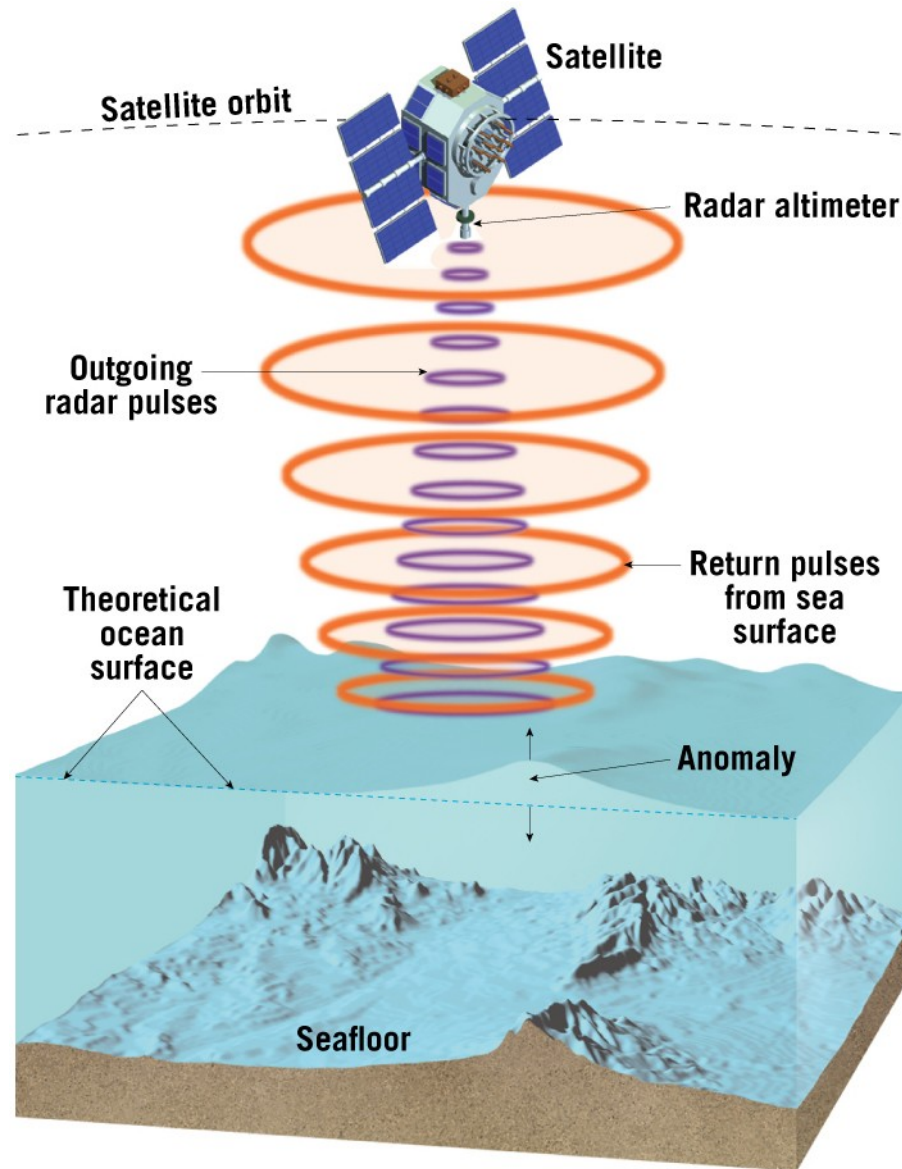
A. Sidescan sonar and multibeam sonar operating from the same research vessel.



10.1 An Emerging Picture of the Ocean Floor

- Mapping the seafloor
 - Mapping the ocean floor from space
 - Massive underwater structures exert stronger-than-normal gravitational attraction.
 - Water piles up over these features.
 - Satellite radar altimeters can detect changes in elevation of the ocean surface.

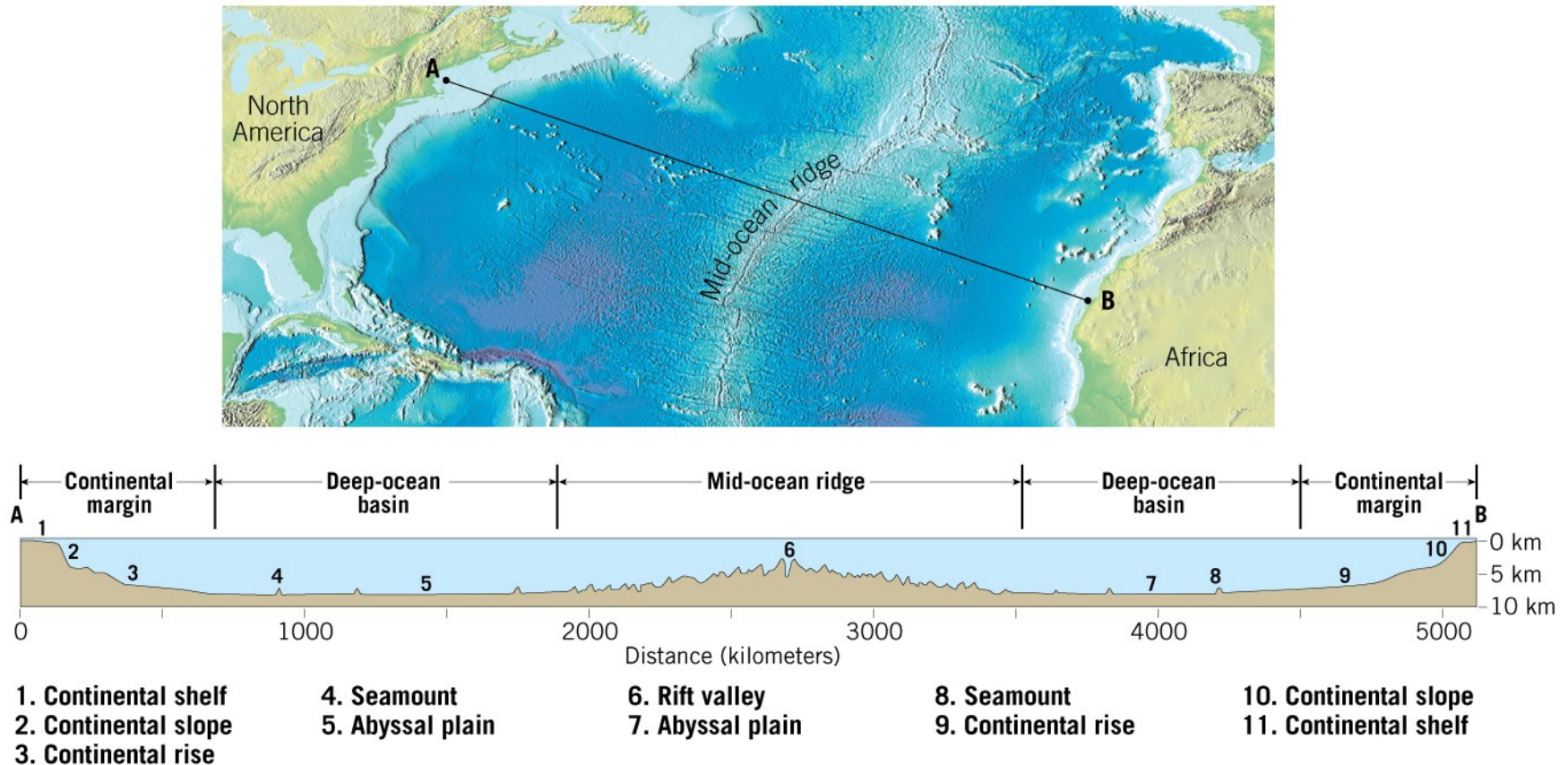
Satellite Altimeter



10.1 An Emerging Picture of the Ocean Floor

- Three major provinces of the ocean floor
 - Continental margins
 - Deep-ocean basins
 - Oceanic (mid-ocean) ridges

Major Topographic Divisions of the North Atlantic



10.2 Continental Margins

- Passive continental margins
 - Found along most coastal areas that surround the Atlantic ocean
 - Not associated with plate boundaries
 - Experience little volcanism and few earthquakes

10.2 Continental Margins (cont.)

- Features comprising a passive continental margin
 - Continental shelf
 - Flooded extension of the continent
 - Varies greatly in width
 - Gently sloping
 - Contains important mineral deposits
 - Some areas are mantled by extensive glacial deposits.

10.2 Continental Margins (cont.)

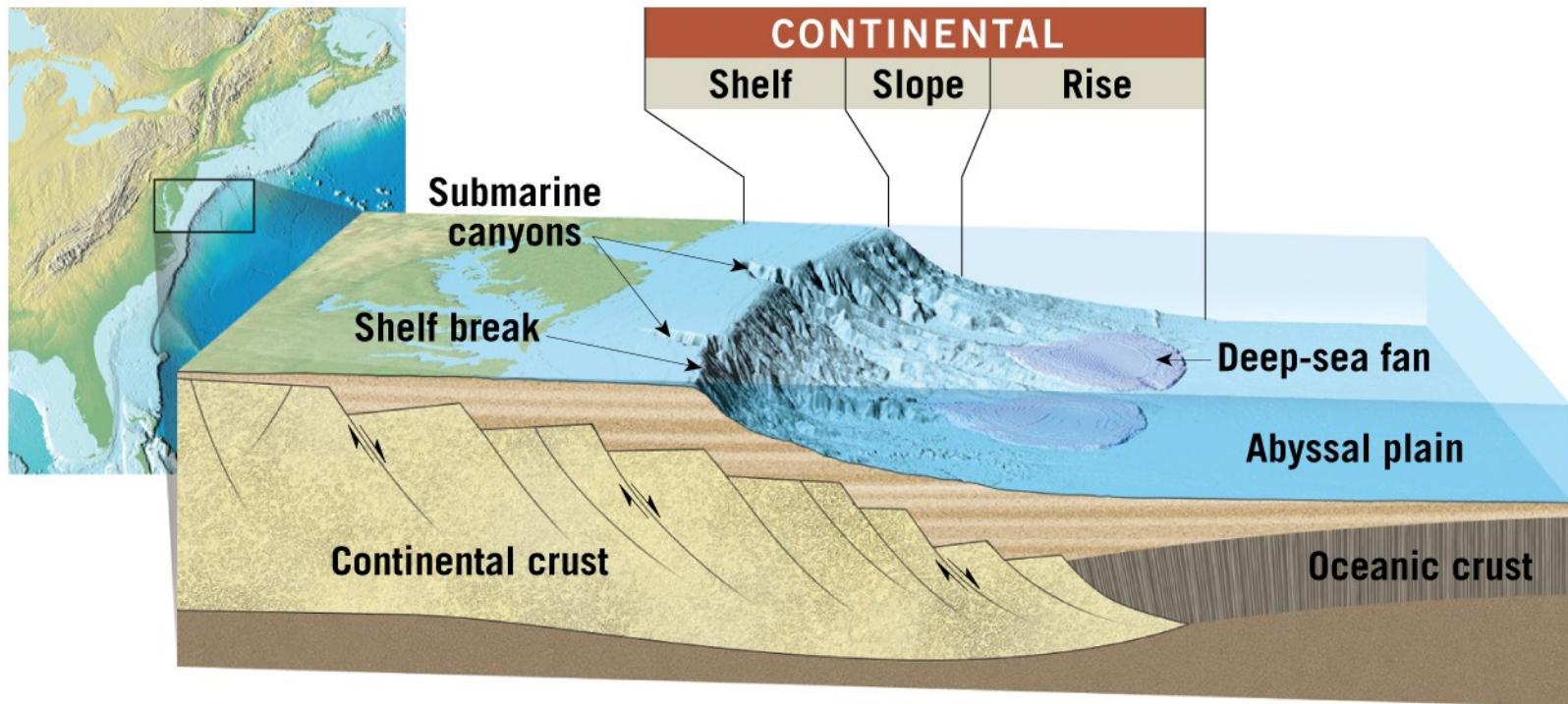
- Features comprising a passive continental margin
 - Continental slope
 - Marks the seaward edge of the continental shelf
 - Relatively steep structure
 - Boundary between continental crust and oceanic crust

10.2 Continental Margins (cont.)

- Features comprising a passive continental margin
 - Continental rise
 - Found in regions where trenches are absent
 - Continental slope merges into a more gradual incline—the continental rise.
 - Thick accumulation of sediment
 - At the base of the continental slope, turbidity currents deposit sediment that forms deep-sea fans.

Passive Continental Margins

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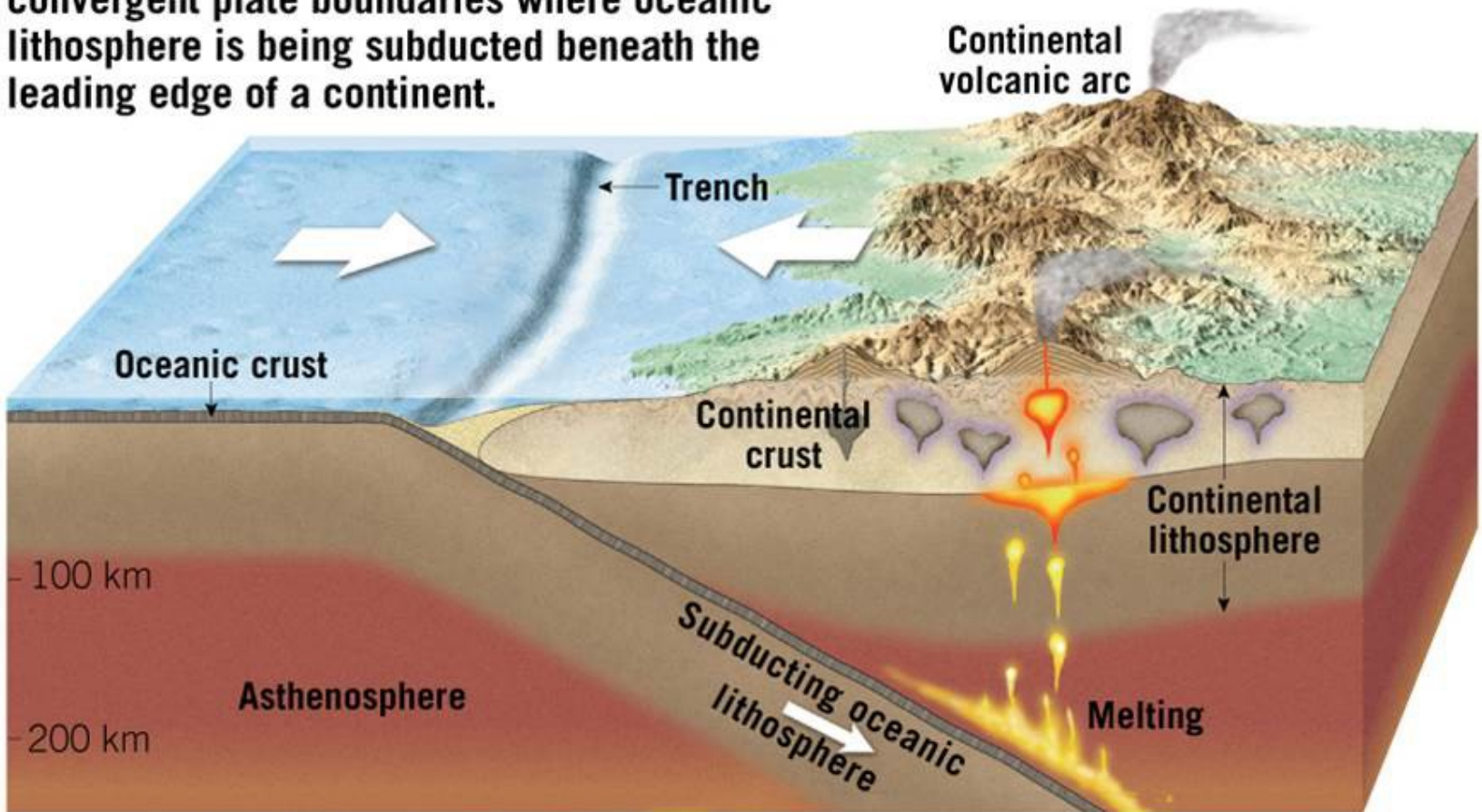


10.2 Continental Margins

- Active continental margins
 - Continental slope descends abruptly into a deep-ocean trench.
 - Located primarily around the Pacific Ocean
 - Accumulations of deformed sediment and scraps of ocean crust form accretionary wedges.

Active Continental Margin

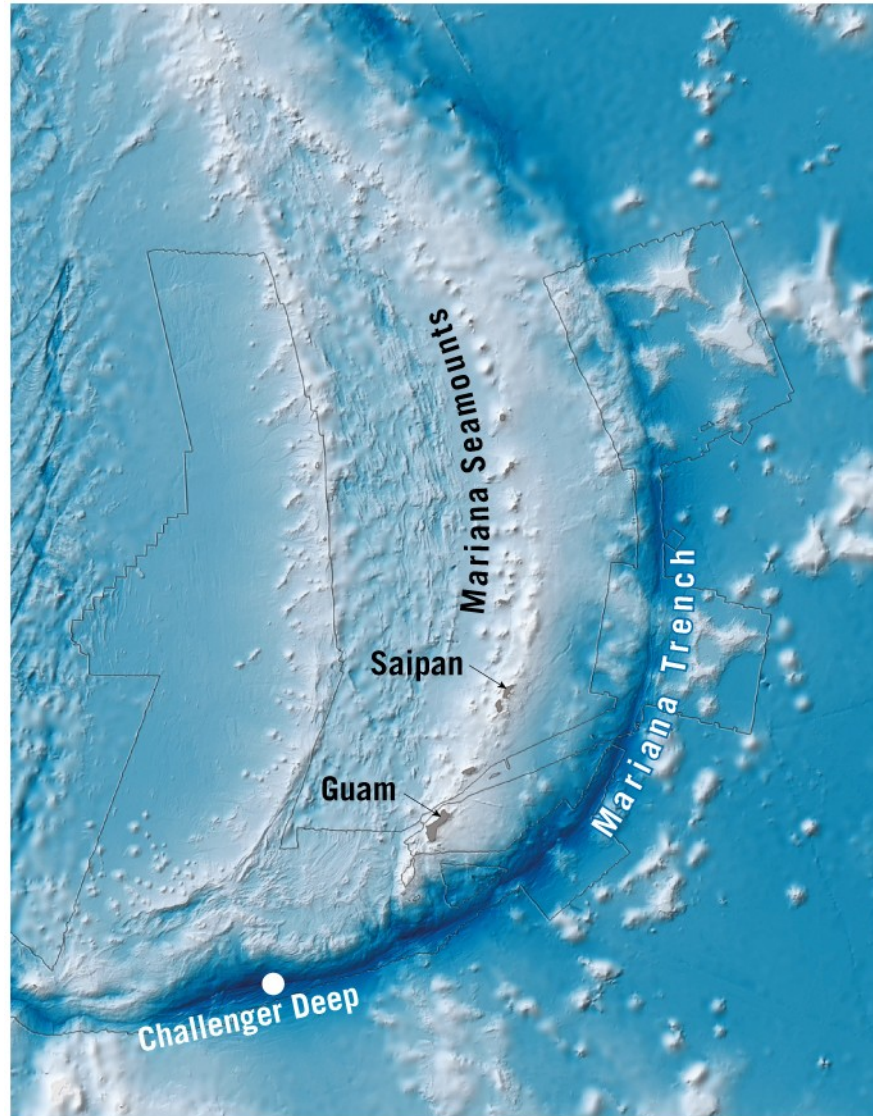
Active continental margins are located along convergent plate boundaries where oceanic lithosphere is being subducted beneath the leading edge of a continent.



10.3 Features of Deep-Ocean Basins

- Deep-ocean trench
 - Long, relatively narrow features
 - Deepest parts of ocean
 - Most are located in the Pacific Ocean
 - Sites where moving lithospheric plates plunge into the mantle
 - Associated with volcanic activity

The Challenger Deep



10.3 Features of Deep-Ocean Basins

- Abyssal plains
 - Likely the most level places on Earth
 - Sites of thick accumulations of sediment
 - Found in all oceans

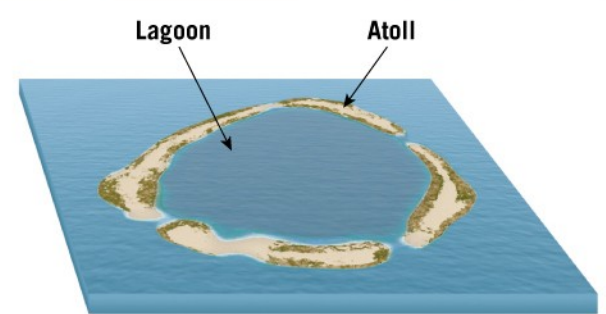
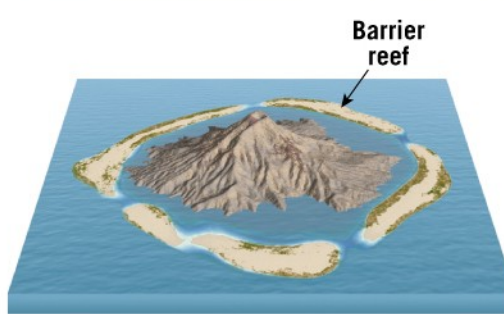
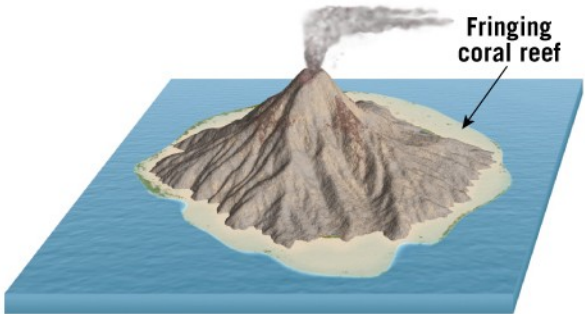
10.3 Features of Deep-Ocean Basins (cont.)

- Volcanic structures on the ocean floor
 - Seamounts
 - Isolated volcanic peaks
 - Many form near oceanic ridges
 - May emerge as an island
 - Guyots
 - Seamounts that sink and form flat-topped mountains
 - Below surface of water
 - Oceanic plateaus
 - Vast outpourings of basaltic lavas on the ocean floor

10.3 Features of Deep-Ocean Basins (cont.)

- Volcanic structures on the ocean floor
 - Coral atolls
 - Ring-shaped structures of corals
 - Found at depths of several thousand meters
 - Darwin hypothesized that this was formed due to subsiding volcanoes.

Darwin's Hypothesis



10.4 Anatomy of the Oceanic Ridge

- Oceanic ridge: Broad, linear swells along divergent plate boundaries
 - Occupy elevated positions
 - Extensive faulting and earthquakes
 - High heat flow
 - Numerous volcanic structures

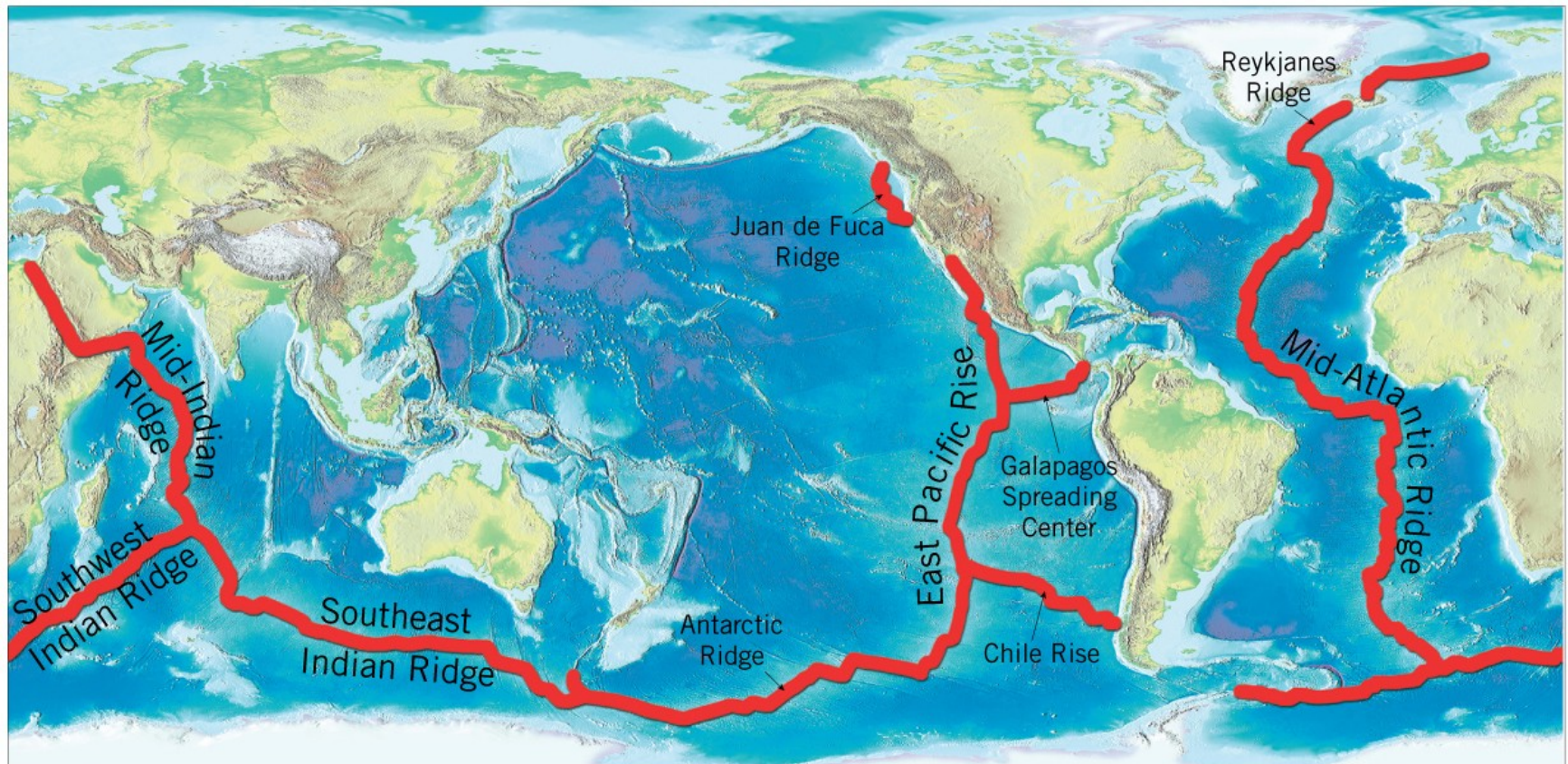
10.4 Anatomy of the Oceanic Ridge (cont.)

- Oceanic ridge characteristics
 - Longest topographic feature on Earth's surface
 - More than 70,000 kilometers (43,000 miles) in length
 - 20% of Earth's surface
 - Winds through all major oceans
 - Term ridge is misleading; widths of 1000 to 4000 kilometers give the appearance of broad, elongated swells

10.4 Anatomy of the Oceanic Ridge (cont.)

- Oceanic ridge characteristics
 - Axis of some ridge segments exhibit deep, down-faulted structures called rift valleys.
 - Portions of the mid-Atlantic ridge have been studied in considerable detail.

Distribution of the Oceanic Ridge System



10.5 Ocean Ridges & Seafloor Spreading

- Seafloor spreading
 - Concept formulated in the early 1960s by Harry Hess
 - Seafloor spreading occurs along relatively narrow zones, called rift zones, located at the crests of ocean ridges.
 - As plates move apart, magma wells up into the newly created fractures and generates new slivers of oceanic lithosphere.
 - New lithosphere moves from the ridge crest in a conveyor-belt fashion.

10.5 Ocean Ridges & Seafloor Spreading (cont.)

- Why are oceanic ridges elevated?
 - Newly created oceanic lithosphere is hot and occupies more volume than cooler rocks.
 - As the basaltic crust travels away from the ridge crest, it is cooled by seawater.
 - As the lithosphere moves away, it thermally contracts and becomes more dense.

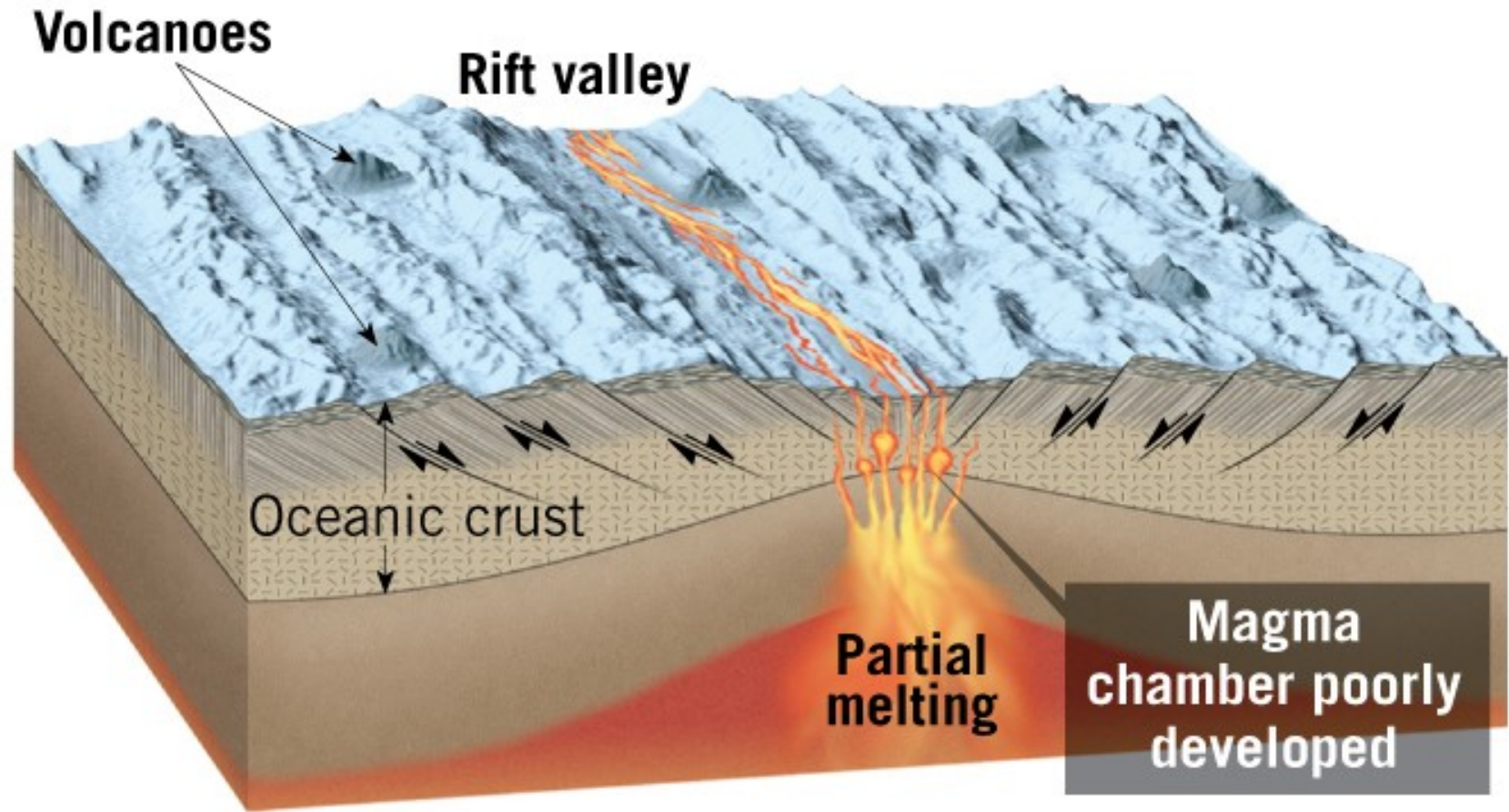
10.5 Ocean Ridges & Seafloor Spreading (cont.)

- Spreading rates and ridge topography
 - Ridge systems exhibit topographic differences.
 - Topographic differences are controlled by spreading rates.
 - At slow spreading rates (1– cm per year), a prominent rift valley develops along the ridge crest that is usually 30 to 50 kilometers across and 1500–3000 meters deep.

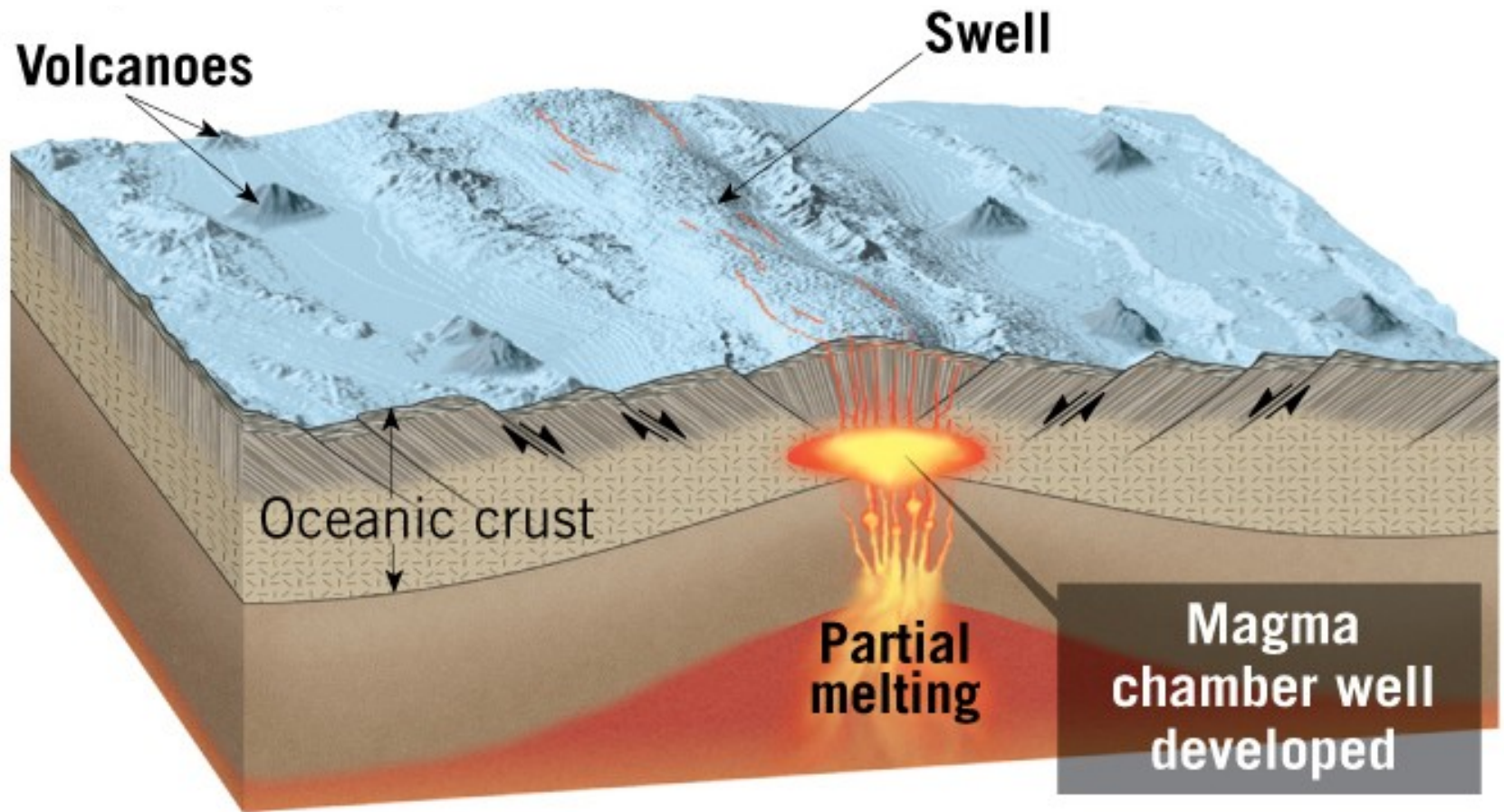
10.5 Ocean Ridges & Seafloor Spreading (cont.)

- Spreading rates and ridge topography
 - Topographic differences are controlled by spreading rates.
 - At intermediate spreading rates (5–9 cm per year), rift valleys that develop are shallow and less than 200 meters deep.
 - At spreading rates greater than 9 centimeters per year, no median rift valley develops and these areas are usually narrow and extensively faulted.

Slow Spreading Oceanic Ridge



Fast Spreading Oceanic Ridge



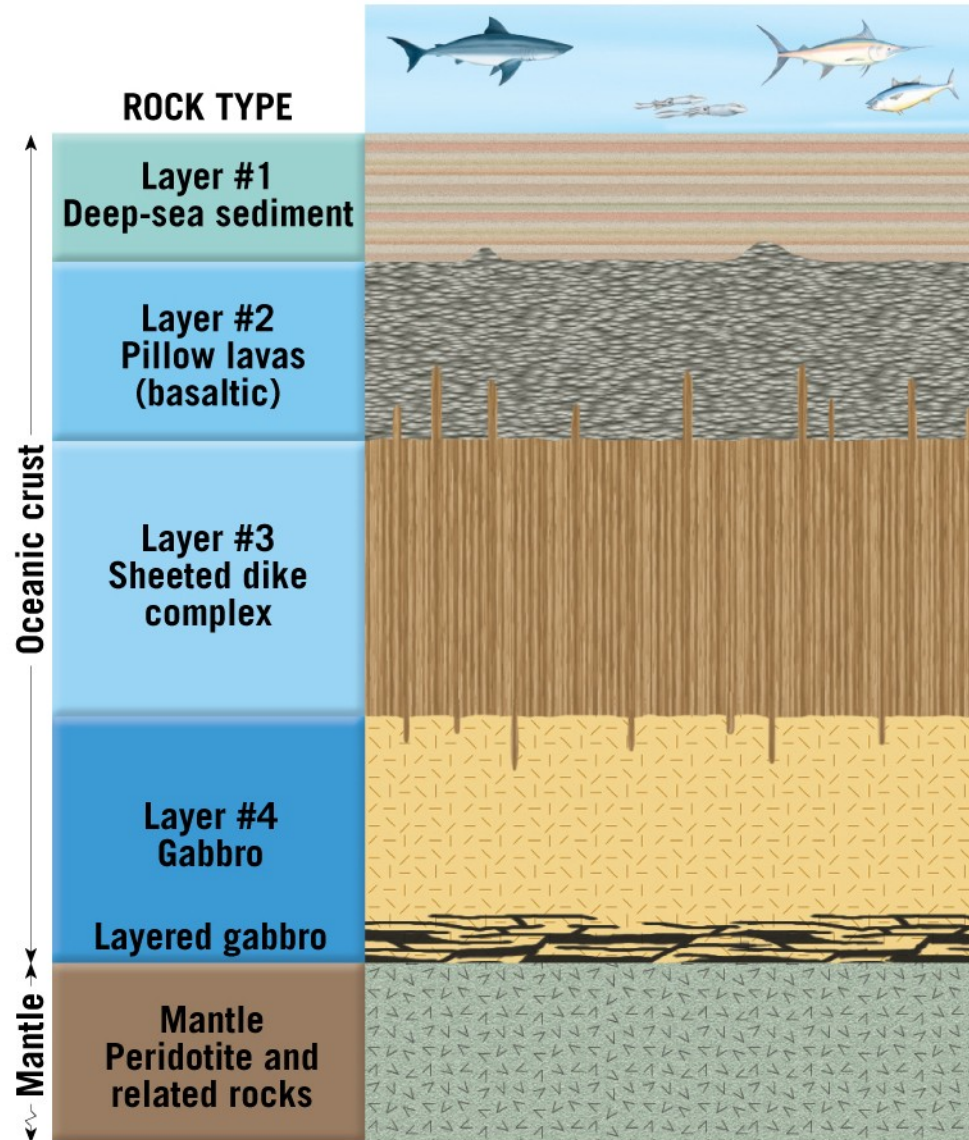
10.6 The Nature of Oceanic Crust

- Four distinct layers, in a sequence of rocks called an ophiolite complex
 - Layer 1: Sequence of unconsolidated sediments
 - Layer 2: Consisting of pillow lavas
 - Layer 3: Numerous interconnected dikes called sheet dikes
 - Layer 4: Gabbro

10.6 The Nature of Oceanic Crust (cont.)

- Formation of oceanic crust
 - Basaltic magma originates from partially melted mantle peridotite.
 - Molten rock injected into fractures above the magma chambers creates the sheeted dike complex.
 - The submarine lava flows chill quickly and the congealed margin is forced forward to produce large tube-shaped protuberances known as pillow basalts.

Structure of the Oceanic Crust



10.6 The Nature of Oceanic Crust

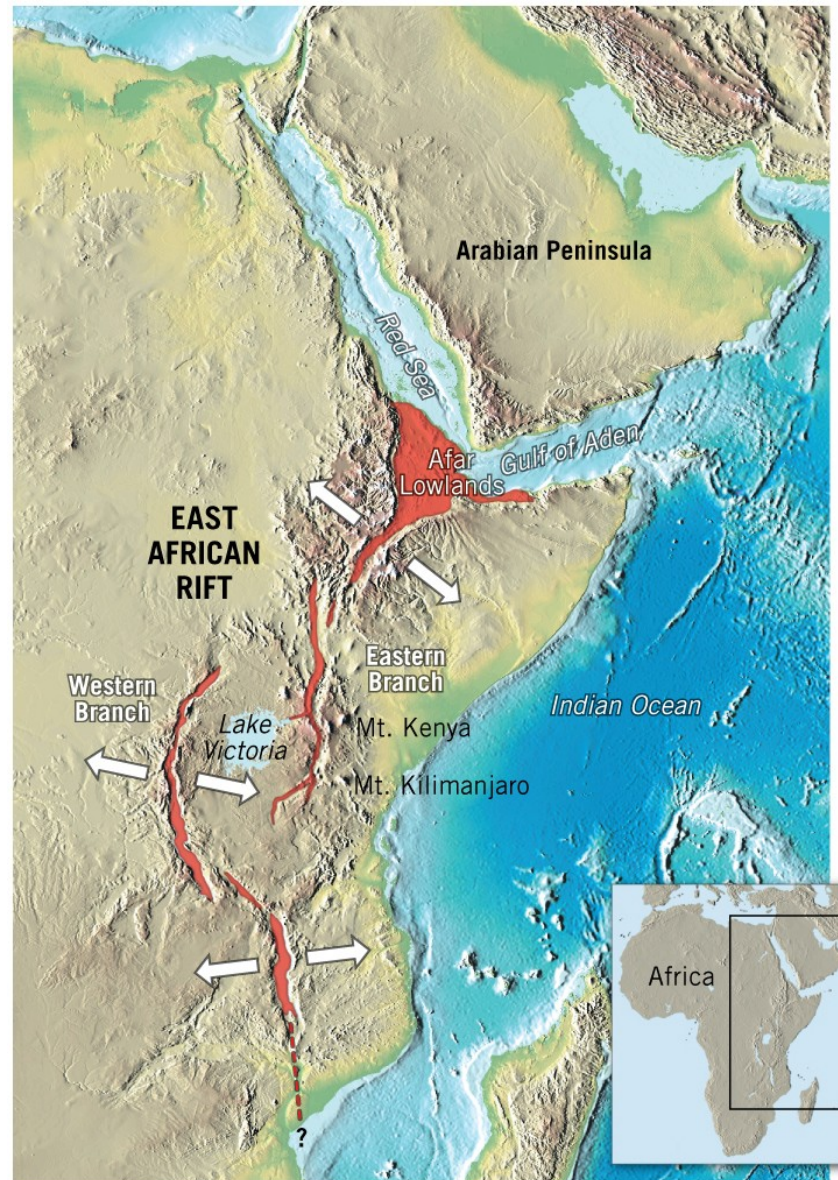
- Interactions between seawater and oceanic crust
 - Seawater circulates downward through the highly fractured crust.
 - Basaltic rock is altered by hydrothermal metamorphism.
 - Hydrothermal fluids dissolve ions of various metals and precipitate them on the seafloor as particle-filled clouds called black smokers.

10.7 Continental Rifting: The Birth of a New Ocean Basin

- Evolution of an ocean basin
 - A new ocean basin begins with the formation of a continental rift.
 - Splits landmasses into two or more smaller segments
 - Examples include the East African Rift, Baikal Rift, the Rhine Valley, Rio Grand Rift, and the Basin and Range.
 - The Red Sea is an example of a rift valley that has lengthened and deepened in a narrow linear sea.

East Africa Rift Valley

Tutorial



10.7 Continental Rifting: The Birth of a New Ocean Basin

- Evolution of an ocean basin
 - Atlantic Ocean
 - After tens of millions of years, the Red Sea will develop into a feature similar to the Atlantic Ocean.
 - As new oceanic crust was added to the diverging plates, the rifted margins moved further from the region of upwelling.
 - These margins cooled and subsided below sea level.

Formation of an Ocean Basin

[Animation](#)

D. Eventually, an expansive ocean basin and ridge system are created.

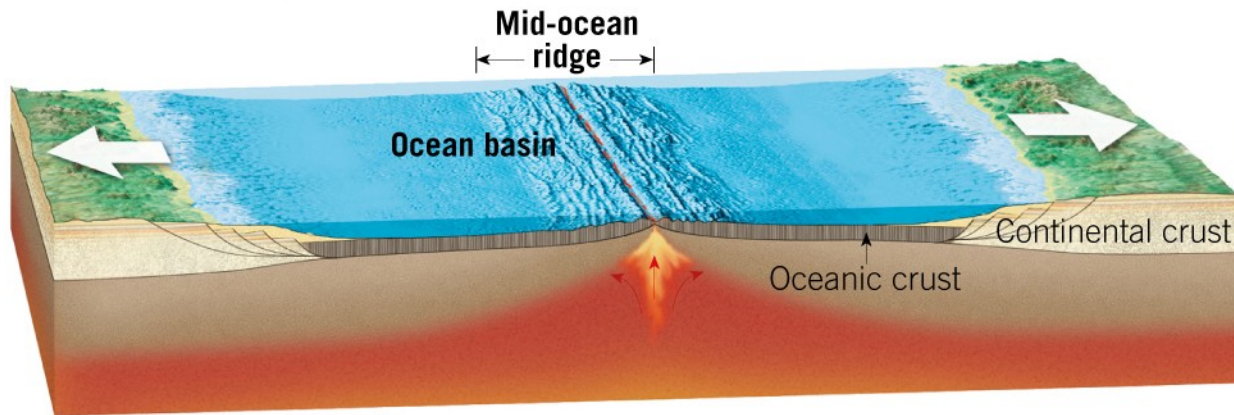
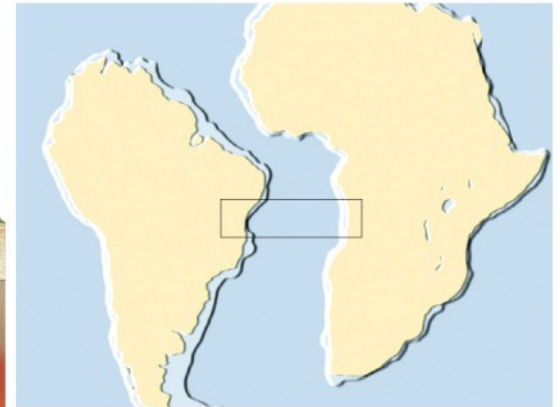


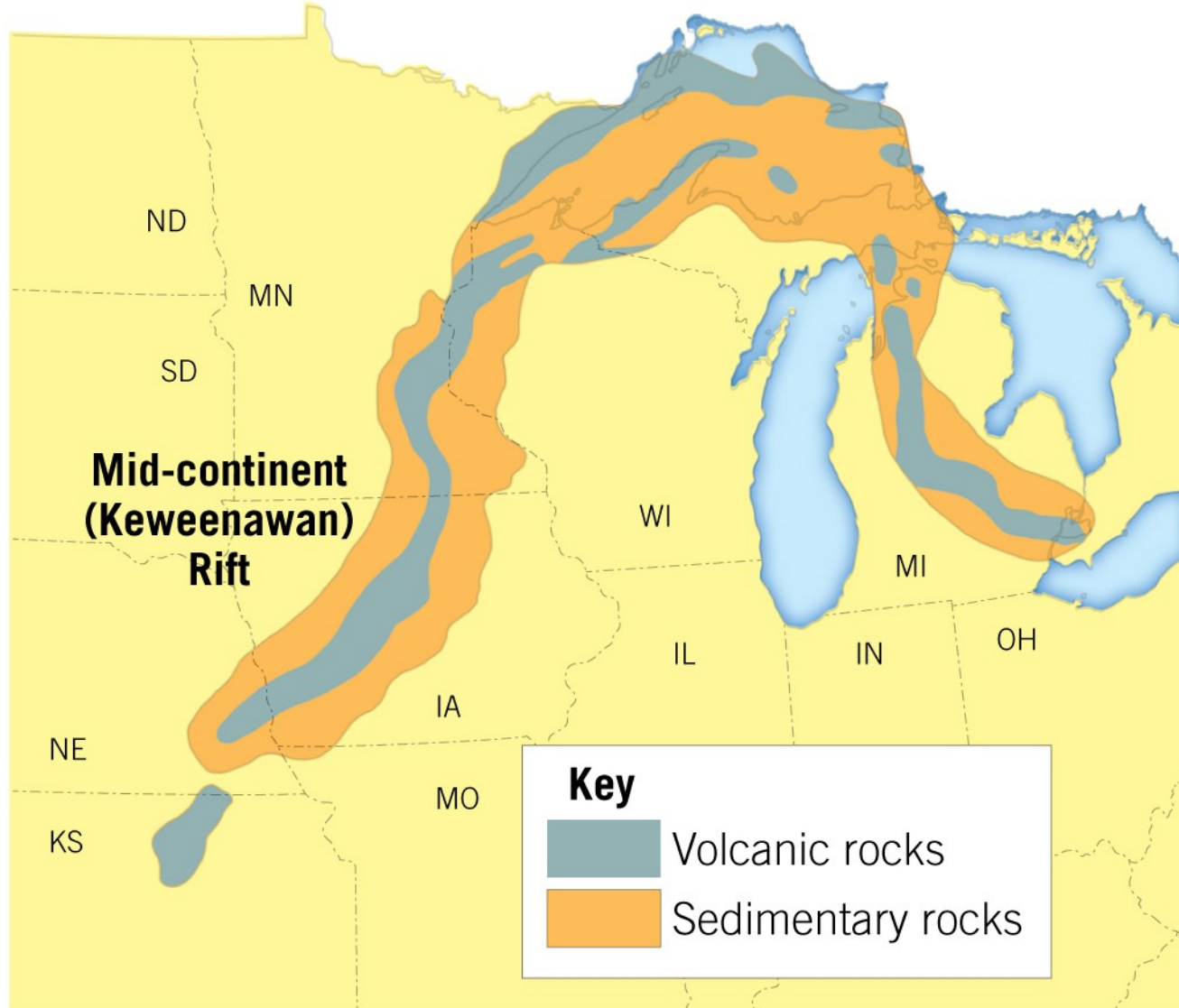
Illustration of the separation of South America and Africa to form the South Atlantic.



10.7 Continental Rifting: The Birth of a New Ocean Basin

- Failed rifts
 - Formerly active rift valley is filled with basalt and clastic sedimentary rocks.
 - Why rifts fail or succeed is not fully understood.

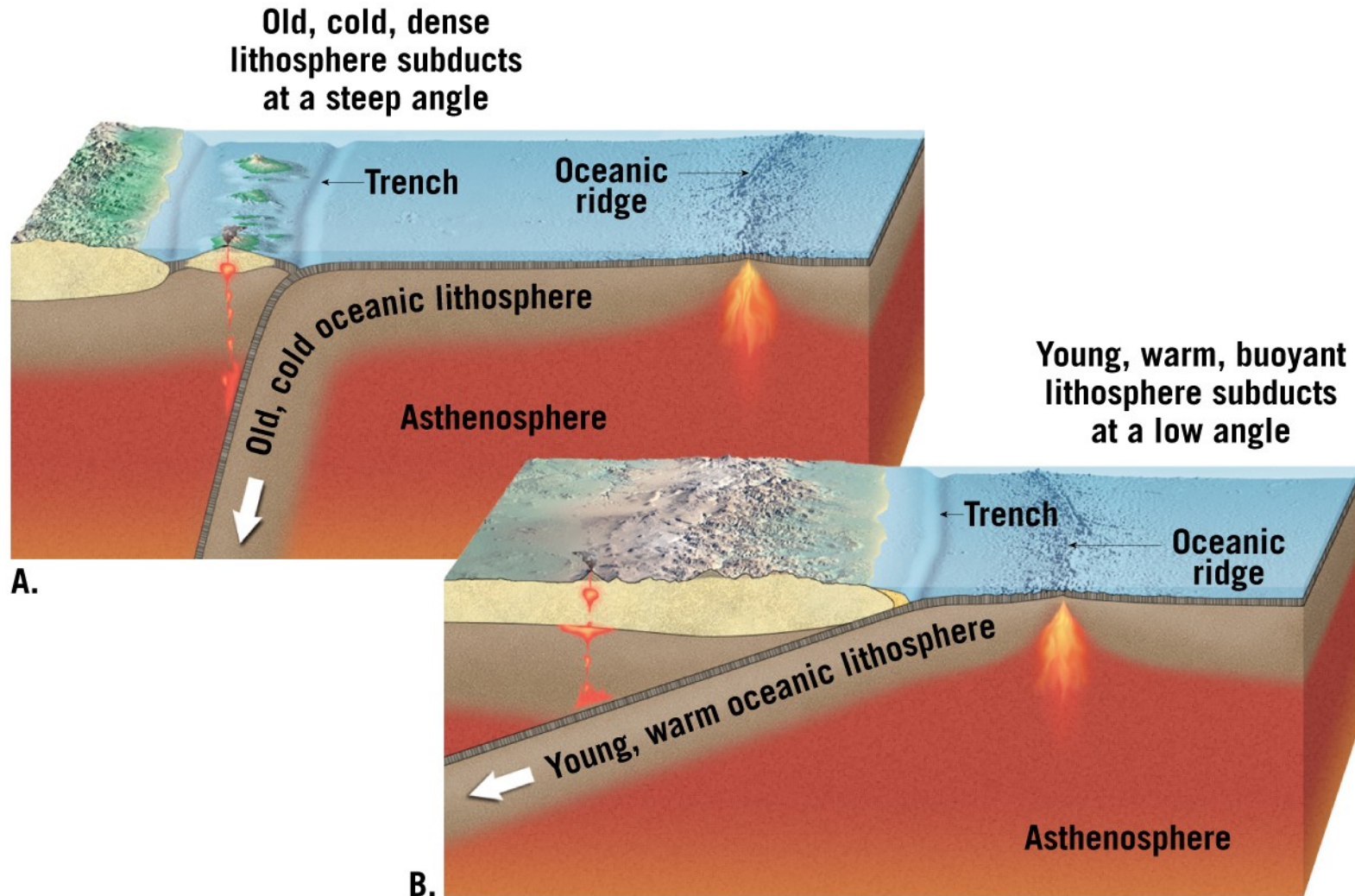
Failed Rift



10.8 Destruction of Oceanic Lithosphere

- Why oceanic lithosphere subducts
 - Spontaneous subduction
 - Very old, thick, dense lithosphere sinks to the mantle by its own weight.
 - Results in descending angles of nearly 90 degrees
 - Example: Mariana trench
 - Lithospheric mantle is what drives subduction.

The Angle of Plate Subduction Depends on Its Density



10.8 Destruction of Oceanic Lithosphere

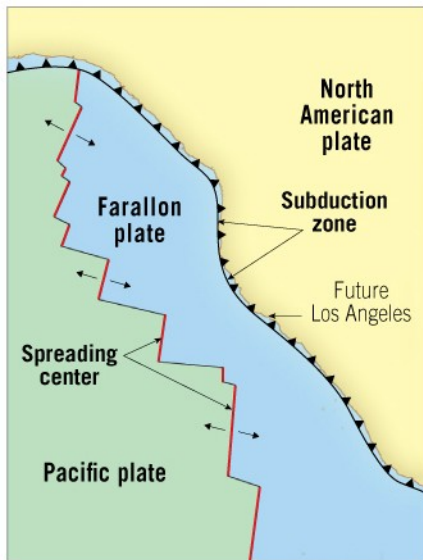
- Why oceanic lithosphere subducts
 - Forced subduction
 - Younger, less dense lithosphere is forced beneath the overlying plate by compressional forces.
 - Descends at shallow angles
 - Example: Peru–Chile trench

10.8 Destruction of Oceanic Lithosphere (cont.)

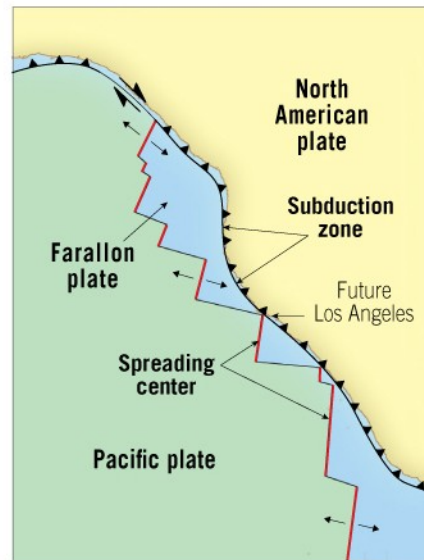
- Subducting plates: The demise of an ocean basin
 - The Farallon plate once occupied much of the eastern Pacific basin
 - Beginning 180 million years ago, the Farallon plate was subducting beneath the Americas faster than it was being generated.
 - The plate got continually smaller and now only fragments of the original plate remain as the Juan de Fuca, Cocos, and Nazca plates.

Demise of the Farallon Plate

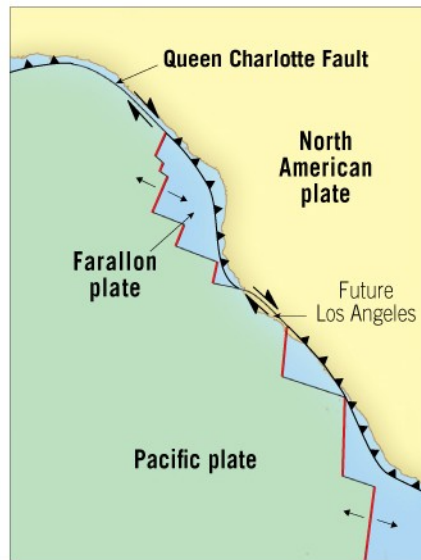
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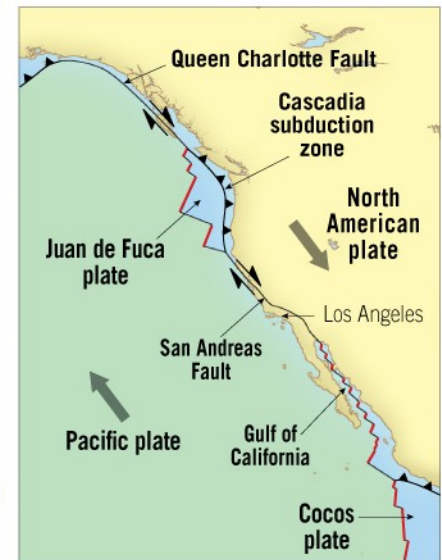
A. 56 million years ago



B. 30 million years ago



C. 20 million years ago



D. Today